

STANLEY CHEUNG

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EDUCATION

The University of British Columbia

B.Sc, Major in Computer Science

September 2022 – May 2027

Vancouver, BC

- **Courses:** Machine Learning, Computer Networking, Relational Databases, Algorithms & Data Structures, Computer Systems & Hardware, Object-Oriented Programming
- **Awards:** 2025 Dean's List, 2024 Charles & Jane Banks Scholarship, 2023 Dean's List

EXPERIENCE

Software Engineer Intern | TypeScript, Express.js, PostgreSQL, AWS, React

VoltSafe Inc.

January 2024 – July 2024

Vancouver, BC

- Designed and implemented an end-to-end role-based access control system with an Express.js, PostgreSQL, and AWS Cognito backend, ensuring controlled access to the internal admin dashboard.
- Developed and Dockerized a server monitoring system using React, Express, and Prometheus, enabling real-time collection and analysis of CPU, RAM, and disk metrics across 7+ EC2 web and database servers.
- Developed an image and document storage system with a RESTful API for the marina management app, using AWS S3 to optimize performance and reduce bandwidth costs and allowing users to upload, store, and retrieve documents and images.
- Conducted a performance analysis of the marina management app using Chrome Lighthouse, reducing total app bundle size by 22% by pruning unused static assets, leading to improved load times.

AP Computer Science Instructor

iSmart Education Canada

October 2022 – February 2023

Vancouver, BC

- Designed and taught weekly CS lessons on algorithms, data structures, and Java programming.
- Exceeded expectations by supporting students, some still in primary school, in achieving 5/5 on the AP CSA 2023 exam.

PROJECTS

UBC Course Explorer | TypeScript, React, Express.js, Node.js | github.com/cheuyin/ubc-course-explorer

- Built a full-stack analytics app for exploring 64,000+ UBC course sections across 189 departments, featuring grade-trend charts, instructor and department profiles, and a campus room finder, using React 19, TanStack Query, and an Express + TypeScript REST API.
- Implemented a JSON query language supporting nested boolean filters (AND/OR/NOT), wildcard string matching, multi-key sorting, and GROUP-BY aggregations (MAX/MIN/AVG/SUM/COUNT), with a validation layer that returns precise field-level error messages; verified by 100+ integration tests.
- Engineered an asynchronous data-ingestion pipeline that unzips raw archives, parses course JSON, and scrapes building/room data from HTML (parse5) into a file-backed store, with job-status polling endpoints for long-running uploads and no external database.

Local AI Coding Agent | Python, Google Gemini API, Rich | github.com/cheuyin/ai-agent-python

- Engineered an autonomous agentic loop using the Google Gemini API, dispatching file system tool calls across up to 20 iterative model rounds until task completion.
- Implemented context window management that tracks prompt token usage per round and trims conversation history when approaching the 1M-token limit, enabling indefinitely long sessions.
- Blocked path traversal attacks via `os.path.commonpath` validation on all file operations; implemented 3-attempt exponential backoff retry for transient API failures.

Pathfinding Visualizer | TypeScript, React, HTML5 Canvas | <https://cheuyin.github.io/pathfinding-visualizer/>

- Built a web app for visualizing pathfinding algorithms (Dijkstra's, A*, and DFS) and random maze-generation algorithms.
- Rewrote the rendering pipeline from React DOM to HTML5 Canvas with `requestAnimationFrame` loops and applied `React.memo/useCallback` memoization, cutting redundant renders 86% (115,000 to 16,000) and raising frame rate 8-10x (7 to 60+ FPS).
- Set up a Continuous Deployment pipeline using GitHub Actions to automate deployment when pushed to main branch.

SKILLS

Languages: TypeScript, Python, SQL, R, Java, C, JavaScript

Frameworks: FastAPI, React, Express.js, SQLAlchemy, LangChain

Other: Node.js, PostgreSQL, Docker, Git, AWS, RESTful APIs, SQL ORMs, Linux, Prompt Engineering

Certifications: AWS Certified Cloud Practitioner